Earth and Planetary Science Letters, 92 (1989) 307–316
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Paleomagnetic study of western Tibet: deformation of a narrow zone along the Indus Zangbo suture between India and Asia

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Received 15 May 1988; revised version received January 23, 1989

More than 150 oriented samples were collected along the Kaxigal-Lhasa road in the southwestern part of the Tibetan Plateau to determine the deformation aspect of the Tibetan region from a paleomagnetic point of view. The counter-clockwise deflection about 60° in declination was revealed at four localities in the narrow zone along the Indus-Zangbo suture zone. A north to northeast direction ($D = -9°$ to $30°$) was observed at four localities, more than 14 km from the suture zone in the western Tibetan region. Significant tectonic deformation has occurred exclusively in a narrow zone whose width is about 40 km spanning the suture zone. The crustal blocks of this narrow zone were rotated in a compressive regime accompanied by left-lateral shear during the collision of the Indo-Eurasia continents. The deformation of the narrow zone behaves as a buffer zone in a tectonically gigantic dimension where the tangential stress is mainly absorbed.

1. Introduction

The Indian continent moved northward since its splitting from Antarctica and collided with the Eurasian continent at about 50 Ma [1,2]. Its collision and subsequent indentation toward the Eurasian continent have given rise to internal deformation in the southern part of Eurasia [3]. Because the Tibetan region has behaved as a bumper against the northward rush of India since 50 Ma [2,4–6], there is abundant geological evidence concerning tectonic deformation preserved in this region. The Tibetan region thus provides a magnificent opportunity to study the deformation of continents.

Paleomagnetism offers a quantitative method for establishing the geological aspect of the deformation of continents. In 1985, during an expedition to the western Tibetan Plateau which followed the 3000 km long Kaxigal-Lhasa road (Fig. 1), more than 150 oriented samples were collected from 22 separate sites for paleomagnetic investigation. We report here the paleomagnetic results, and discuss the tectonic deformation of the southern part of the Tibetan Plateau.

2. Geology and sampling

Western Tibet is divided into five geologic provinces: from south to north are the Himalaya, Gamnian Qiangtang, Kunlun and Tarim provinces [7].

Ophiolite sequences appear along the Yarlung-Zangbo river [8,9], which is called Indus-Zangbo suture zone. This marks the boundary between the Himalaya province to the south and the other four provinces to the north. This suture zone represents the site of the Paleogene collision between the Indian and Eurasian continental plates.

The Himalaya province is mainly composed of a suite of marine deposits of the period between Precambrian and Cenozoic. The fauna of this province contains the typical Tethyan forms. This indicates that the sedimentary sequence of this province was formed on the oceanic crust of the

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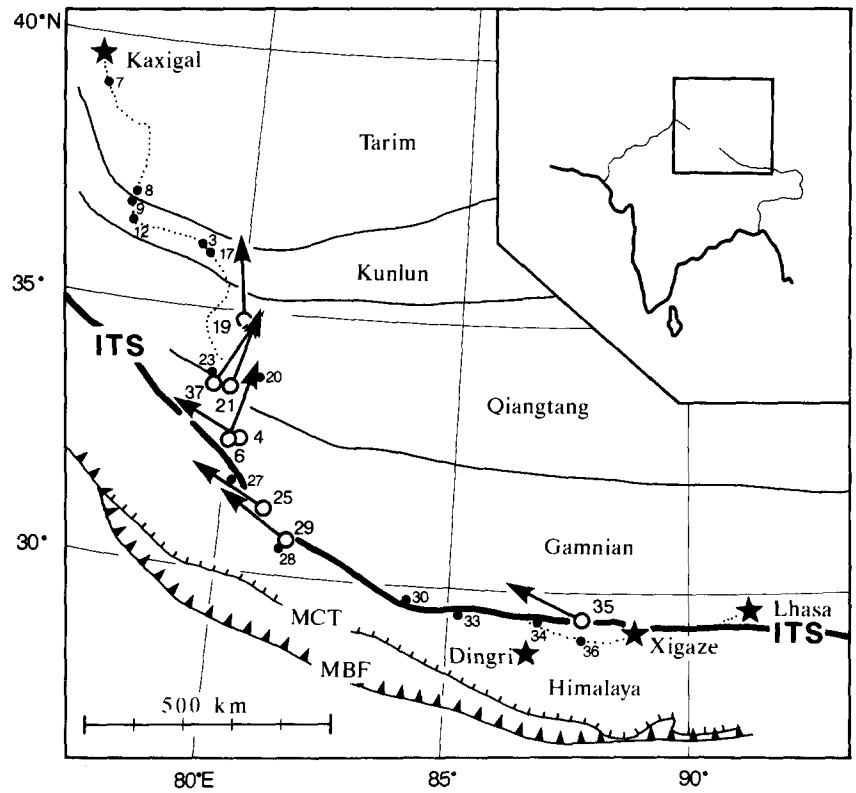


Fig. 1. Sketch map of the major tectonic units and the geologic provinces in western Tibet. Heavy line is the Indus-Zangbo suture zone (ITS), and barbed lines are continental thrusts (*MCT* = Main Central Thrust; *MBF* = Main Boundary Fault). Sampling localities are indicated by closed and open circles along the sampling route (dotted line) from Kaxigal to Lhasa. Numerals by circles are site numbers. Arrows indicate the declinations of the reliable paleomagnetic directions obtained from eight sites (drawn by open circles).

Tethyan Ocean between the Indian and the Eurasian continents [10].

The Gamnian province is composed of Lower Paleozoic to Mesozoic sediments with intercalated volcanic rocks. The southeastern portion of this province is dominated by the east–west trending calc-alkaline Gandese batholith and its associated volcanic sequence in which radiometric ages range between 120 and 40 Ma [11–16]. Most of the Gandese Belt is attributed to magmatism at an active continental margin above a northward-dipping subduction zone [17]. This province formed the southern margin of the Eurasian continent prior to about 40 Ma.

The northern three provinces have occupied the interior of the Eurasian continent since the Mesozoic. The Qiangtang province is marked by fluvial materials of clastics, gravels and red bed sequences. The oldest rocks are Permian, which include the typical of the Cataysian flora (*Gigantopteris*) [18]. The Kunlun province is chiefly composed of the Carboniferous to Triassic marine sediments of turbidites, mudstone, slates and limestone. Lower Carboniferous carbonates–clastics yield Laurasian coral and brachiopod faunas [18]. The Tarim province is characterized by the extensive veil of the Quaternary sands. They hide the Precambrian to Tertiary sequences.

We collected samples from 22 sites (Fig. 1). Twelve sites were from the Himalaya and Gamnian provinces along the suture zone. The site positions from these provinces are located within 23 km from the suture zone. The remaining sites were from the other three provinces, and are more than 40 km north of the suture zone. The details are given in the appendix for each site locality and its stratigraphic relationship. Topographic maps (O.N.C) at 1 : 1,000,000 and the geological map at

1 : 1,500,000 [19] were used to locate the site positions.

Hand samples were taken and a magnetic compass was used for orientation. Each site typically comprised five independently oriented samples, distributed over distances ranging up to 50 m. The present declination value at each sampling point was estimated from the calculation using the International Geomagnetic Reference Frame of 1985 [20].

3. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from the samples. The specimens were measured on a two-component ScT cryogenic magnetometer. The stability of natural remanent magnetization (NRM) was investigated by either thermal or alternating field (AF) method. Specimens were thermally demagnetized using a laboratory-built furnace. The residual magnetic field in which the specimens were cooled was less than 10 nT. AF demagnetization was carried out with a three-axis tumbler system in one cylindrical mu-metal shield. Preliminary demagnetization experiments for pilot specimens indicate that the thermal demagnetization method was effective for erasing the secondary magnetization. All the remaining samples were progressively demagnetized in steps of 30–50 °C using the thermal method.

Zijderveld component plots [21] and equal area projections were used to assess the directional stability and unblocking temperature spectrum for each specimen. Demagnetization results were evaluated by visual inspection followed by principal component analysis [22]. Specimen directions with a maximum angular deviation (MAD) [22] less than 10° were accepted to be reliable ones.

3.1. Demagnetization

Treatment in this analysis revealed a large variety of behavior among samples during demagnetization. Sampling sites were divided into two categories according to the results of thermal demagnetization of the samples: category A—samples showed similar behavior in each site, and category B—samples showed a large variety of behavior even within one site.

3.2. Category A

This category comprises twelve sites, which are further divided into three groups according to a number of components of the NRM: single, double and multi-component.

Single component (sites 19, 36, 37). Only three sites have samples with a single magnetic component. The unblocking temperature varies from site to site (Fig. 2). Samples of site 37 revealed a sudden decrease in intensity between 550° and 600 °C without any directional change. The NRM is probably carried by titanium-poor magnetite. Although the NRM of the samples of site 19 showed monotonous convergence to the origin in the orthogonal plots, the NRM was carried by two kinds of magnetic carriers whose unblocking temperatures were 300 °C and 500 °C. More than 30% of the NRM of site 36 remained after demagnetization to 600 °C, indicating that hematite was a carrier of the NRM. Although each sample of site 36 showed the monotonous demagnetization, their directions were not clustered.

Double component (site 4, 6, 7, 8, 28, 29, 34, 35). Examples of orthogonal demagnetograms are shown in Fig. 2. Almost all samples had a low-temperature component which was removed in the initial stages of demagnetization up to treatment of 200 °C. Further thermal treatment revealed only one high-temperature component with a demagnetization trajectory that converged toward the origin. The unblocking temperature of the high-temperature component varied from 250° to 700 °C. The low-temperature component is most likely a viscous remanent magnetization acquired in the recent geomagnetic field. The characteristic directions were obtained for these sites from the high-temperature component after treatment of demagnetization at above 100 °C.

Multi component (site 20). Three components were observed in one site (Fig. 2). The lowest-temperature component was removed by 100 °C. This component is probably ascribable to the low-temperature component observed in the group with double component. The medium-temperature component was removed by 250 °C. The high-temperature component was identified above

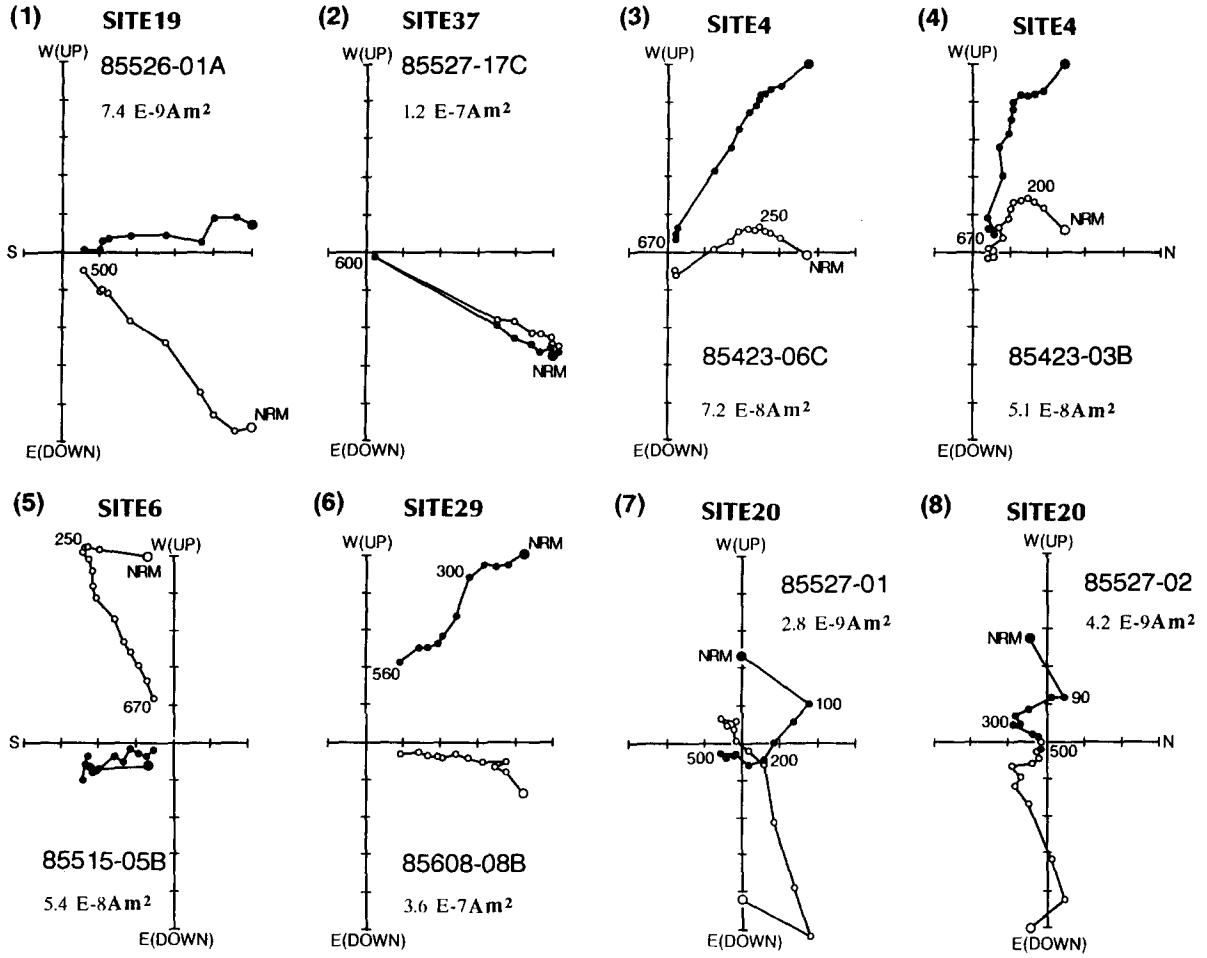


Fig. 2. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens of category A (see text). (1), (2): specimens with a single magnetic component from site 19 and 37. (3), (4), (5), (6): specimens with two magnetic components from site 4, 6 and 29. (7), (8): specimens with three magnetic components from site 20. Open (solid) symbols show the magnetic vectors projection on to the vertical (horizontal) plane. Numbers by symbols represent temperature (°C). Tectonic tilt corrections have not been applied.

350°C and it is recognized as the characteristic direction for this site.

3.3. Category B (site 3, 9, 12, 17, 21, 23, 25, 27, 30, 33)

The samples of the remaining ten sites had variable components within one site; single, double and triple in the NRM directions. However, almost all samples of eight sites in the ten sites had stable end points. The characteristic directions of these sites were identified in the higher-temperature component except for site 25. The low-temperature component is, in general, ascribable to a viscous remanent magnetization.

A typical example is drawn in Fig. 3. Half of the samples from site 21 had two components, while the remaining samples showed three components. The highest-temperature component for these samples revealed a reversed NRM direction, indicating that the characteristic direction appeared after demagnetization at above 350°C.

Samples from site 25 consisted of two types (Fig. 3): (1) samples with a stable, single component direction, which showed northwest declination with steep inclination; and (2) samples with more than two magnetization components. The low-temperature component was observed up to 350°C and its direction revealed northwest

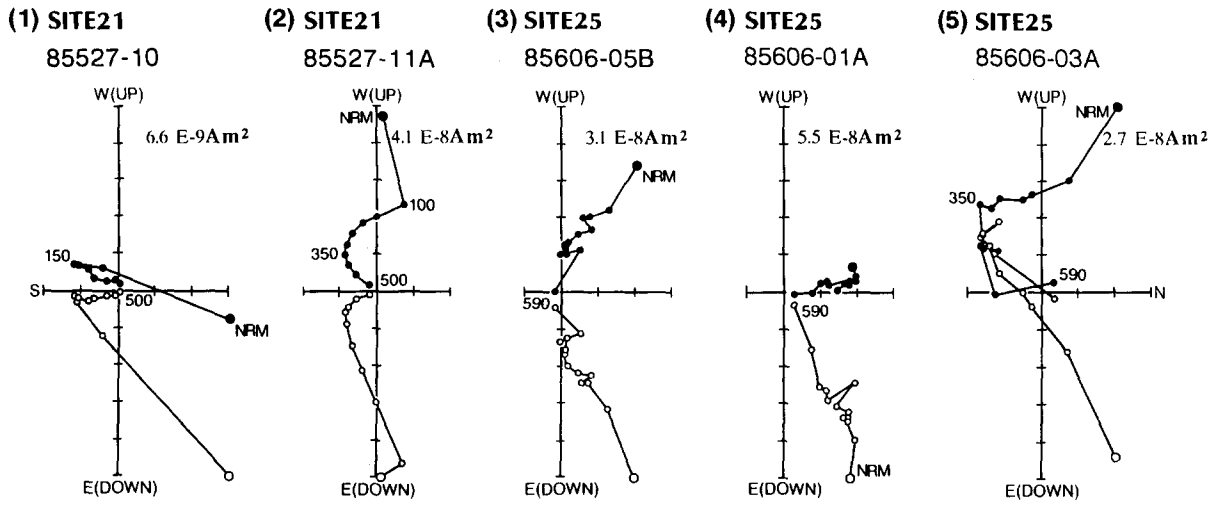


Fig. 3. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens of category B (see text). (1), (2): site 21 comprises samples with two or three magnetic components. (3), (4), (5): site 25 comprises samples with single or multi-magnetization component. Open (solid) symbols show the magnetic vectors projection on to the vertical (horizontal) plane. Numbers by symbols represent temperature (° C). Tectonic tilt corrections have not been applied.

declination value. The higher-temperature components had variable directions among samples. The samples demagnetized at above 350 ° C revealed a gradual decrease in the magnitude of their NRMs within the field free space inside the cryogenic magnetometer: it took more than 25 minutes for the NRM direction of some specimens to reach a stable end point. This is probably due to minera-

TABLE 1

Paleomagnetic results

Site	Geological time	Demag. level range (° C)	N	(n)	Before tilt correction				After tilt correction				Distance from ITS (km)
					D (°)	I (°)	α_{95} (°)	K	D (°)	I (°)	α_{95} (°)	K	
7 *	Tp	100–610	6	9	10.0	46.4	13.2	16.1	7.8	47.3	13.7	15.2	240
8 *	S–D	100–500	5	5	9.1	58.9	14.1	30.3	70.5	41.2	17.6	20.0	150
19	K	250	5	7	−9.7	32.9	11.0	31.1	−8.5	−10.4	11.9	26.6	225
20 *	J	85–500	6	6	−12.3	56.2	17.2	16.1	−25.9	11.5	21.5	10.7	165
21	J–K	300–500	4	5	−161.1	11.3	21.6	13.5	−162.5	−28.2	23.1	11.9	85
37	K	450	5	10	30.1	24.3	3.3	220.7					40
6	Ter	150–670	6	9	−178.6	−40.1	9.7	29.1	−164.6	−22.7	12.0	19.3	14
4	K	150–670	6	11	−65.3	−17.5	6.0	58.3					5
25	Tpl	0–400	5	10	−44.2	56.2	18.9	7.5	−61.5	52.0	18.9	7.5	11
29	K	100–560	3	11	−61.1	6.8	22.8	5.0					0
35	K–Tpl	450	5	11	−90.4	32.9	29.1	3.4	−66.9	−2.6	27.9	3.6	23

N, n are number of samples and specimens. A mean direction for each site is calculated by assigning unit weight to specimens. A mean direction is computed after Kirschvink [22] in the range of unblocking temperature where the characteristic direction has been identified. Since site 19, 35 and 37 have a narrow range in unblocking temperature, the Kirschvink's method is not available. The mean direction for these sites is computed at a temperature at which the characteristic direction has been identified.

S = Silurian; D = Devonian; J = Jurassic; K = Cretaceous; Ter = Tertiary; Tpl = Paleogene; Tp = Pliocene.

The distance from ITS is measured from Indus or Yarlung-Zangbo rivers. The ITS is ambiguous at a region between sources of these rivers, because more than two lines of ophiolite sequences appear. The distance of sites 4 and 6 is measured from the nearest line of ophiolite sequences.

Data denoted by asterisk are parallel to the present geomagnetic field or the axial dipole field before tectonic correction. No paleomagnetic investigation was conducted for these sites.

logical modification and not due to the magnetization process. Magnetization acquired by the newly formed magnetic minerals is possibly responsible for divergence of NRM directions of higher-temperature components. Because the NRM directions of the samples of this site clustered tightly during the initial stages of demagnetization, and their northwest direction is not parallel to the present geomagnetic field before tilt correction, the low-temperature component is probably a characteristic direction of this site.

No stable-end points were identified for any samples from two sites (site 17 and 33) during demagnetization. The different demagnetization behavior among samples is probably due to the weakness of the NRM intensity. The NRM directions of the samples began a random walk after their intensity decreased to less than 10^{-5} A/m. No stable remanent magnetization could be isolated from these samples.

4. Results

In 19 sites could the characteristic directions with the MAD less than 10° be isolated. Mean direction for each site was calculated by assigning unit weight to a specimen. We arbitrarily discarded results from eight sites for which the circle of 95% confidence exceeded 30° in order to exclude poorly determined and spurious directions form the data set. Thus, eleven sites remained (Table 1).

Three sites (sites 7, 8, 20) appear to record a secondary remanent magnetization. Their characteristic directions show normal polarity, parallel to the present geomagnetic field or the axial dipole field before tectonic correction. The rocks possibly acquired their remanent magnetization after tilting, in some recent normal polarity epoch, probably the Brunhes chron. These sites are excluded from further paleomagnetic consideration.

The characteristic remanent directions of the remaining eight sites were corrected for simple tectonic tilt where possible. Their declination values are plotted in Fig. 1. Only post-Jurassic rocks preserved the reliable paleomagnetic directions.

The paleomagnetic directions are divided into two categories according to the declination value (Fig. 4):

(A) Westward deflection of about 60° is observed in two sites of sedimentary rocks along the suture zone both before and after tilt correction. The remanent magnetization of granite and gabbro from the suture zone also reveals a large amount of westward declination of about 60°, although no tilt correction is available. Inclination values are distributed in fairly shallow range except for site 25.

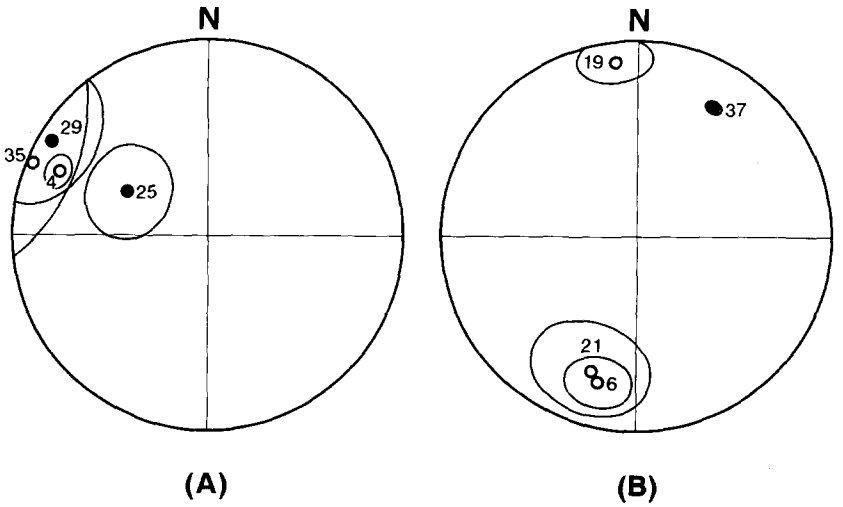


Fig. 4. Mean magnetic directions and 95% confidence circles. A. Paleomagnetic data from sampling sites close to the Indus-Zangbo suture zone (0–23 km from the suture). B. Paleomagnetic data from sampling sites fairly apart from the Indus-Zangbo suture zone (14–225 km from the suture). Site number refers to Table 1. Projections are equal area, solid (open) symbols on the lower (upper) hemisphere.

(B) A north to northeastward direction ($D = -9°$ to $30°$) is observed at four sites which are well away from the suture zone. These sites also revealed shallow inclination ($-10°$ to $28°$). The reliability of the northward direction is ascertained by the presence of both normal and reversed polarities.

5. Discussion

5.1. Shallow inclination and northward movement

We observe low inclinations for the post-Jurassic rocks in western Tibet. The present inclination values of this area are expected from the IGRF of 1985 [20]. The expected inclinations are between 44° and 52°, whereas the observed ones are between $-18°$ and 28° except for the site 25 ($I = 52°$). The observed inclinations are more than 20° shallower than the expected ones.The inclination values expected from Eurasian paleomagnetic pole of Cretaceous of 100 Ma [23] are between 30° and 40° in western Tibet. Comparison with the paleo-inclination of 100 Ma indicates that the observed inclinations are still shallower than the expected paleo-inclinations.

The observed inclination of western Tibet is fairly concordant with the data of Cretaceous to Paleogene rocks reported from the North and South Lhasa blocks [24, 25] and the Tethyan Himalaya [6]: their inclination values are distributed between $-6°$ and 41°. This concordance strongly suggests that the rocks of western Tibet acquired their remanent magnetization at a low latitude, probably between equator and 20°N in paleolatitude near where the Lhasa blocks and the Tethyan Himalaya were situated in the Cretaceous and the Paleogene [24]. We conclude that western Tibet also underwent northward translation since the Paleogene.

5.2. Counter-clockwise paleomagnetic rotation

The most striking result from this work is a large counter-clockwise deflection in the paleomagnetic direction in the narrow zone along the Indus-Zangpo suture zone (Fig. 1). The amount of the deflection is well within the small range between $-61°$ and $-67°$. The deflection is observed along about 500 km of the suture zone; the westernmost site (site 4) with westward deflection of declination is 82°20′E in longitude and the easternmost site (site 37) is 87°41′E.

A counter-clockwise deflection in declination ($D = -85 \pm 20°$) was reported from the Cretaceous Xigaze ophiolite and flysh in the central part of the Indus-Zangbo suture zone [26] (Fig. 1). This declination value is similar to the westward declination ($D = -61°$ to $-67°$) observed in western Tibet along the suture. The easternmost site with westward declination extends to 89°20′E.

This counter-clockwise deflected declination is in contrast with the northward direction observed at the locations away from the suture zone in the Tibetan and Himalaya regions; western Tibet, the Lhasa blocks [24] and the northwest Himalayas [25]. The large amount of counter-clockwise deflection appears to be an intrinsic character of remanent magnetization for a narrow zone along the suture. The narrow zone exceeds probably 700 km in length.

5.3. Width of the narrow zone with counter-clockwise declination

Site 35 is located 23 km to the north of the ophiolite sequence at the Indus-Zangbo suture. This site is the most distant sampling locality from the suture with the westward deflected direction in western Tibet. The Xigaze Group in central Tibet, with a large counter-clockwise declination, is less than 20 km north of the suture [26]. The counter-clockwise declination is also observed on the south of the suture: site 4 is 5 km south of the ophiolite outcrop in western Tibet. The narrow zone characterized by the counter-clockwise rotation extends up to about 20 km on either side of the suture.

All sampling sites more than 50 km away of the suture show a northward declination in western Tibet. No large counter-clockwise paleomagnetic rotation is reported from either side of the Indus-Zangbo suture: The Lhasa blocks and Dingri area which are more than 50 km away from the suture (Fig. 1) exibit declination values between $-27°$ and $-6°$ [6,24,25]. The narrow zone does not, therefore, exceed 100 km in width across the suture. The width of the narrow zone is probably less than 40 km.

5.4. Deformation of the narrow zone

The paleomagnetic data strongly suggest that a significant paleomagnetic rotation occurred exclusively within a narrow zone along the Indus-

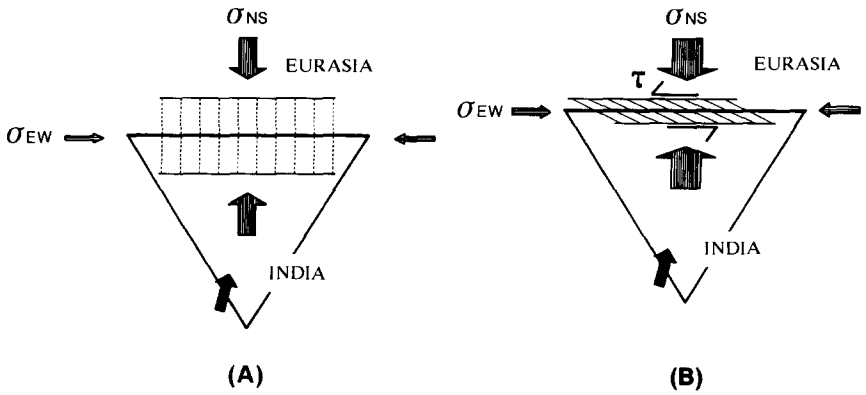


Fig. 5. Domino-style deformation of the narrow zone between the Indian and Eurasian continents. A. When the normal stress on the suture zone (σ_{NS}) became larger than the east–west normal stress (σ_{EW}) during collision of India with Eurasia, the north–south direction (dotted lines) was the orientation of incipient slip planes for crustal blocks. B. As the north–south normal stress (σ_{NS}) increased up to the critical state, the crustal blocks began to slide along the slip planes (solid lines) at their contact. The left-lateral shear (τ) set in the domino-style deformation with lateral displacement to the left in the narrow zone between the Indian and Eurasian continents. Black arrows: direction of motion of the Indian continent relative to Eurasia.

Zangbo suture in the Tibetan region. The counter-clockwise deflection is not attributed to the geomagnetic behavior (e.g. excursion and secular variation), because the age of samples spans a long period between the Cretaceous and the Paleogene. Tectonic rotation due to deformation is a plausible explanation for the paleomagnetic rotation within the narrow zone.

Timing of the deformation is inferred to be later than the Cretaceous or Paleogene on the basis of the age of the rocks with the westward paleomagnetic rotation. The Indian and Eurasian continents collided with each other since the Paleogene [2,6], and the Indus-Zangbo suture zone was a front of the collision. The contemporaneousness of the tectonic events between the collision and the deformation along the suture zone suggests that the collision is responsible for the counter-clockwise paleomagnetic rotation in the suture zone.

This tectonic rotation is explained by a large-scale tectonic manifestation of the "domino style" deformation (Fig. 5). This kinematics of the deformation arises under the condition of compressive regime accompanied with left lateral shear [27]. The collision between India and Eurasia gave rise to the left-lateral shear (τ) as well as the intensive north–south compressive stress (σ_{NS}) along the Indus-Zangbo suture zone [10,28–30]. The left-lateral shear is expected from the eastward sense in motion of India suggested by the geological study of Indoburman region [28–30]. Domino-style deformation accounts for a constant amount of the paleomagnetic rotation over a long range along the Indus-Zangbo suture zone.

The "ball bearing model" [31] may be an alternative mechanism for the deformation of the crustal blocks along the strike-slip fault. However, it is hard to explain the constant amount of the paleomagnetic rotation with this mechanism, because each block moves independently in the ball bearing model. Geological structures in the Tibetan region [10,32] indicate that few north–south faults, separating the independent crustal blocks, can be observed across the suture: east–west faults prevail along the Indus-Zangbo suture zone and the sedimentary formation along the Indus-Zangbo suture zone seems to be stretched along the east–west direction. These geological observations suggest that the deformation is not due to the rotation of independent blocks but due to domino style deformation of the crustal blocks across the suture zone.

The narrow, deformed zone along the Indus-Zangbo suture probably behaves as a mechanical buffer zone of shear stress in the compressive regime, because the domino style deformation effectively consumes tangential stress. Geological observation [17,18,33] suggests that folding, thrust faulting and crustal thickening in the Tibetan Plateau and Himalaya region absorb normal stresses acting on the suture zone of continental

boundary between India and Eurasia, whereas little tangential shear stress propagates deep into the interior of either continent. Shear stress was presumably accommodated in the narrow deformed zone along the suture during collision tectonics.

Acknowledgements

We would like to acknowledge the continuous encouragement of Professor Kazuo Huzita. T. Nishiyama and X. Zheng are indebted to the members of the China-Japan Joint Friendship Expedition to Naimona'nyi 1985 for the courtesy during field survey. Anonymous referees critically read this paper and provided useful suggestions. A part of the present research has been supported by Grant-in-Aids for Overseas Field Research of the Japanese Ministry of Education, Science and Culture, to which we are very much indebted.

Appendix 1

The following information is listed for each site in which a reliable paleomagnetic direction was obtained: site number, site location, rock type, and stratigraphical relation after geological map of the Tibetan Plateau [19]. Sites are listed from north to south.

Site 7 (38°55′N, 76°17′E). 9 km east of Yingjisha. Pliocene silt stone. The Atushi Group in the Shache subprovince of the Tarimu geologic province.

Site 8 (37°09′N, 77°03′E). 26 km east of Sukuluke. Silurian to Devonian limestone. The Sukuluke Group in the Tiekelike subprovince of the Tarimu geologic province.

Site 19 (34°47′N, 80°06′E). 8 km northwest of Quan-Shuihu. Cretaceous siltstone. The Tielongtan Group in the Kala Kunlun subprovince of the Qiangtang geologic province.

Site 20 (33°40′N, 80°27′E). 7 km southeast of Duoma. Jurassic shale and sandstone. The Qiangtang subprovince of the Qiangtang geologic province.

Site 21 (33°31′N, 79°54′E). On the east coast of the Bangongcuo. Jurassic to Cretaceous shale. The Jinzhushan Group in the Qiangtang subprovince of the Qiangtang geologic province.

Site 37 (33°20′N, 79°43′E). 12 km southeast of Ritu. Cretaceous granite. The Qiangtang geologic province.

Site 6 (32°30′N, 80°03′E). Shiquanhe. Tertiary red shale. The Qilin subprovince of the Gamnian geologic province.

Site 4 (32°33′N, 80°22′E). 12 km southwest of Jiangba. Cretaceous quartz diorite. The Gamnian geologic province.

Site 25 (31°16′N, 80°56′E). 14 km northeast of Mentu. Paleogene sandstone. The Zongbai Group in the Shenzha subprovince of the Gamnian geologic province.

Site 29 (30°43′N, 81°22′E). On the south coast of the Laangcuo. Cretaceous gabbro. The Gamnian geologic province.

Site 35 (29°25′N, 87°41′E). Yamu. Cretaceous to Paleogene sandstone. The Xigaze Group in the Xigaze subprovince of the Himalaya geologic province.

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